VIStream: Bringing Inertial Sensing to Streaming Geometric Foundation Models

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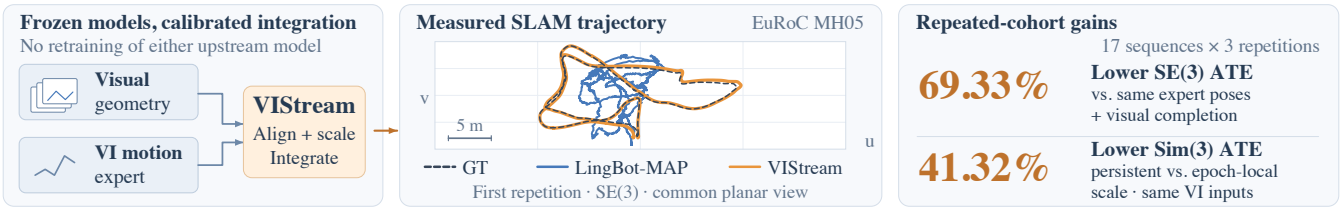


Fig. 1. VIStream at a glance. Frozen-model motion integration, an MH05 trajectory (first repetition; shared-support SE(3); common planar view), and repeated-cohort gains under the indicated controls.

***Abstract*— Streaming geometric foundation models have recently shown strong capabilities for online 3D perception, but their camera motion remains purely visual and lacks a persistent metric reference. We introduce VIStream, an inertial streaming geometric foundation model that augments frozen streaming visual geometry with metric visual–inertial motion. The key challenge is not simply to add inertial estimates, but to integrate them consistently across coordinate frames, scale, tracking interruptions, and estimator resets. VIStream therefore calibrates visual and inertial motion into a common camera frame, maintains metric scale as a persistent physical state, selectively integrates valid rotation and translation updates, and falls back to visual motion when inertial estimates are unavailable. Across 17 EuRoC and TUM-VI sequences, VIStream reduces trajectory ATE by 64.97% over the frozen visual model. Controlled experiments show that persistent metric scale is critical: keeping scale fixed across tracking epochs reduces ATE by 41.32% compared with re-estimating scale after each reset. More importantly, even when using exactly the same visual–inertial estimator outputs, VIStream achieves 69.33% lower SE(3) ATE and 91.00% lower translation RPE than directly using the estimator poses with the same visual fallback.**

***Index Terms*— SLAM, visual–inertial odometry, motion integration, visual foundation models**

I. Introduction

Accurate camera motion is central to SLAM [1]. DUSt3R, VGGT and LingBot-Map [2], [3], [4] encode visual geometric priors. VIStream combines frozen visual geometry with an existing visual–inertial estimator, aligning camera frames, timestamps and length units. Persistent scale, component-wise admission and visual continuation produce the camera trajectory; frozen-depth placement is auxiliary. Fig. 1 summarizes the integration and measured SLAM benefits.

Our contributions are:

- Calibrated SLAM motion integration with frozen upstream models, component-wise admission and visual continuation.
- Persistent-scale integration across expert epochs, evaluated using the same inertial outputs and support.
- Repeated matched-support trajectory and local-motion evaluation with explicit alignment, coverage and complementary controls.

Current tracking uses the expert with loop closure disabled. Historical loop-off/on results remain separate context.

II. Related Work

Visual–inertial estimation and calibration. MSCKF [5] provides a filtering formulation; OpenVINS [6] offers a filter-based platform. On-manifold preintegration [7] provides efficient inertial constraints. OKVIS, VINS-Mono, and ORB-SLAM3 [8], [9], [10] combine visual and inertial observations through nonlinear estimation. Initialization and spatial-temporal calibration [11] affect validity. Our adapter retains ORB-SLAM3's estimator; expert-only, completion, and map controls distinguish expert quality from interface effects. **Geometry priors and learned tracking.** DUSt3R [2] predicts pointmaps, MASt3R [12] grounds matching in 3D, and VGGT [3] predicts cameras and geometry. DROID-SLAM and DPVO [13], [14] combine learned correspondence updates with differentiable bundle adjustment. MASt3R-SLAM [15] integrates reconstruction priors into dense SLAM; VGGT-SLAM [16] optimizes projective submap transformations on $SL(4)$. CUT3R, LingBot-Map, and HorizonStream [17], [4], [18] retain sequential context. HorizonStream already scales depth and translation together. VIStream studies external visual–inertial motion; shared scaling and visual representation are not claimed as novelties.

3D representations in SLAM. KinectFusion, ElasticFusion and BundleFusion [19], [20], [21] couple tracking with volumetric or surfel maps and pose-dependent surface integration. iMAP, NICE-SLAM and Co-SLAM [22], [23], [24] use implicit scene representations for joint tracking and mapping. Gaussian Splatting SLAM and SplaTAM [25], [26] maintain Gaussian scenes for pose estimation and rendering. Kimera [27] couples visual–inertial estimation with metric-semantic meshes; NGEL-SLAM [28] combines feature tracking and neural submaps. VIStream retains frozen

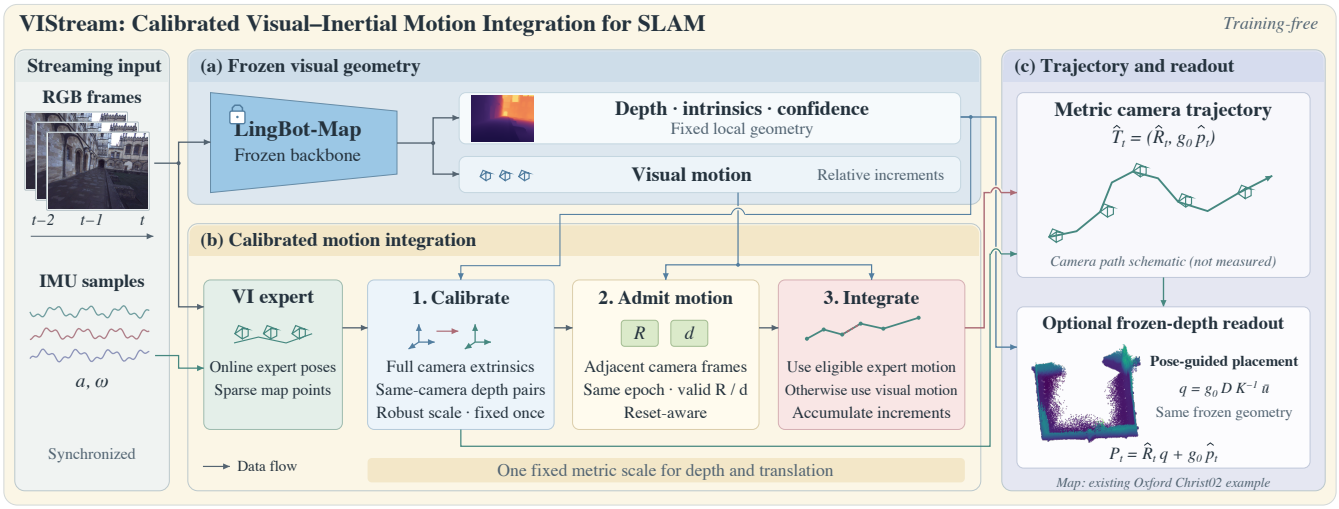


Fig. 2. VIStream integrates camera motion from frozen visual geometry and an existing visual–inertial expert. Calibration, valid-component admission and visual continuation produce a metric camera trajectory; shared g_0 supports an optional frozen-depth map readout. Oxford Christ02 images, depth and map are measured examples; the camera path and motion icons are schematic.

local geometry and evaluates pose-dependent placement as an auxiliary SLAM readout.

III. Calibrated Motion Integration for SLAM

A. Complementary camera-motion estimates

At RGB arrival t, the frozen visual model supplies camera-to-world pose $B_t = [R_t^v, p_t^v]$, camera-axis depth D_t^v, intrinsics K_t and confidence. The existing visual–inertial estimator, ORB-SLAM3 in our implementation, supplies metric camera motion from images and IMU. VIStream maintains a camera trajectory by composing eligible expert increments with visual continuation when an expert connection is unavailable.

The upstream models remain unchanged. The expert supplies tracking and map management; depth supports shared metric calibration and auxiliary 3D placement.

B. Physical alignment across modalities

The monocular-inertial expert processes the same images and IMU measurements with fixed timing calibration. It infers metric velocity and pose from visual observations, gyroscope angular velocity and accelerometer specific force. Motion refers to the same calibrated camera center. With T_{AB} mapping coordinates from B to A, an expert body pose is converted using

$$T^e_{WC,t} = T^e_{WB,t} T_{BC}. \tag{1}$$

The translation in T_{BC} retains the rotating lever arm. Calibration is independent of GT evaluation alignment.

Length conversion uses same-camera depth correspondences. Stable expert points with positive camera-axis depths z^e_{tj} sample visual depths at calibrated pixels u_{tj}, accounting for resize, crop and rectification. Every contributing bilinear corner must have finite positive depth and confidence at least 1.5. With at least 32 supported points, define

$$\ell_{tj} = \log D^v_t(u_{tj}) - \log z^e_{tj}, \qquad s_t = \exp(\mathrm{median}_j\, \ell_{tj}). \tag{2}$$

An initialized, valid expert observation is accepted when the median absolute log deviation is at most $\log 1.5$. Its first accepted value s_0 converts expert lengths to the visual gauge; $g_0 = 1/s_0$ converts visual lengths to metres. The correspondence mask does not select points by depth-ratio agreement.

The scale remains fixed, assuming a fresh SI-calibrated inertial map, no imported atlas, and disabled loop closure. The expert resolves its inertial initialization variables [29]; VIStream does not re-estimate gravity, velocity or bias. Resets clear motion adjacency and validity while retaining the metre convention. Initialization need not give accurate scale; arbitrary monocular unit changes violate this assumption.

C. Validity-aware camera-motion integration

Let $\Delta B_t = B_{t-1}^{-1} B_t = [R_v, d_v]$ and let a valid expert increment be $\Delta E_t = [R_e, d_e]$. Copied outputs from the current tracking call must connect adjacent camera arrivals within one map epoch. A verified service increment is preferred; differencing returned poses is allowed only when their revision identifiers agree. Missing endpoints, resets, or unsafe snapshots prevent the connection. Reacquisition starts a new adjacent pair.

Rotation is eligible at an angular rate of at most $8\,\mathrm{rad\,s^{-1}}$. Translation additionally requires the fixed scale and $\|d_e\|/(z^e_{\mathrm{med}}\Delta t) \le 5\,\mathrm{s^{-1}}$, using at least five stable positive expert depths for z^e_{med}. The primary method adopts eligible components at unit weight:

$$R^\Delta_t = \begin{cases} R_e & \text{eligible rotation,} \\ R_v & \text{otherwise,} \end{cases}$$
$$d^\Delta_t = \begin{cases} s_0 d_e & \text{eligible translation,} \\ d_v & \text{otherwise,} \end{cases} \tag{3}$$
$$\widehat{B}_t = \widehat{B}_{t-1}[R^\Delta_t, d^\Delta_t], \quad \widehat{B}_0 = B_0.$$

Here $[R, d]$ is a homogeneous rigid transform. Rotation may remain eligible when translation is rejected. Visual increments

bridge interruptions. Cumulative integration neither bounds drift nor re-anchors to absolute expert poses.

D. Relation to absolute expert poses

Let $X_t = [\widehat{R}_t, g_0\widehat{p}_t]$ be the metric output. On one contiguous block with fixed scale and fully admitted increments from a coherent expert trajectory E_t, rigid composition [30] gives

$$X_t = X_a \prod_{k=a+1}^{t} (E_{k-1}^{-1}E_k) = H_a E_t, \qquad H_a = X_a E_a^{-1}. \tag{4}$$

The time-ordered product yields $X_i^{-1}X_j = E_i^{-1}E_j$ within this block. A rigid camera-center fit on identical block timestamps absorbs H_a, giving the same optimal SE(3) ATE.

Visual continuation or component mixing can introduce different offsets across blocks; one whole-route fit need not absorb them. Direct placement restores its original expert anchor. Native increments inconsistent with returned-pose revisions also violate the premise. This identity explains the evaluation distinction; jointly varying admission, completion and revision histories does not establish component-level causality.

E. Sequential camera poses and 3D output

After scale initialization at t_0, the adapted metric pose is $\widehat{T}_t = [\widehat{R}_t, g_0\widehat{p}_t]$. The first eight visual outputs wait until input eight; later estimates use the current arrival. First outputs bind pose, depth, scale and times, separately from retrospective revisions. Pre-t_0 outputs remain in native units without metric backfill. Throughput is evaluated separately from sensor rate.

The same pose and scale place frozen local geometry into the scene:

$$P_t(u) = \widehat{R}_t\big(g_0 D_t^v(u) K_t^{-1}\bar{u}\big) + g_0\widehat{p}_t. \tag{5}$$

Here $\bar{u}$ is a homogeneous image pixel. This 3D readout fixes depth and appearance, so its changes reflect pose-dependent placement. It provides a secondary geometric check on the SLAM trajectory; the full projection protocol and pose-sensitivity bound are retained in the Supplementary Material.

IV. Experiments

A. Questions and study design

The evaluation separates system accuracy, interface attribution and sensor-mode effects. Each control states what changes, what remains fixed and which published timestamps are scored. Full trajectories assess accumulated error; adjacent motion tests local consistency. Frozen-depth mapping is secondary.

Current motion interface. Three Oxford Spires [31] development sequences—Christ Church 02, Keble College 04, and Observatory Quarter 01—each contain 3,840 RGB/IMU camera arrivals. Checkpoint, calibration, unit-weight motion rule, depth processing and primary depth factor one are fixed. The current monocular-inertial expert runs with loop closure disabled; SLAM tracking is supplied by the existing estimator.

TABLE I

Full repeated SLAM study: first-output Sim(3) ATE [m]. N is common GT-supported frames per repetition; all three repetitions are retained. VIStream shows mean ± sample SD; the frozen visual result is constant across repetitions. Bold: lower original mean.

Sequence	N	LingBot-Map	VIStream
EuRoC			
MH_01_easy	3638	2.54	**0.23** ± 0.02
MH_02_easy	2999	2.74	**0.51** ± 0.41
MH_03_medium	2631	2.91	**0.34** ± 0.09
MH_04_difficult	1976	5.70	**1.26** ± 1.66
MH_05_difficult	2221	5.68	**0.24** ± 0.08
V1_01_easy	2871	1.61	**0.27** ± 0.03
V1_02_medium	1671	1.64	**1.58** ± 0.00
V1_03_difficult	2094	1.46	**1.09** ± 0.47
V2_01_easy	2240	1.79	**0.31** ± 0.03
V2_02_medium	2309	1.83	**0.62** ± 0.09
V2_03_difficult	1890	1.77	**0.49** ± 0.11
TUM-VI			
Room 1	2764	**0.97**	0.98 ± 0.05
Room 2	2591	0.99	**0.87** ± 0.38
Room 3	2552	**0.74**	1.20 ± 0.03
Room 4	2187	1.07	**0.54** ± 0.19
Room 5	2837	0.93	**0.91** ± 0.00
Room 6	2544	0.92	**0.91** ± 0.01
EuRoC mean	–	2.70	**0.63**
TUM-VI mean	–	0.94	**0.90**
Overall mean	–	2.08	**0.73**

We report all three completed runs using actual first outputs of Eq. 3.

Retained historical experiments. The earlier batch implementation aligns completed trajectories and completes expert boundaries. We retain KITTI 00–10, ten Oxford, three TUM-VI rooms, and EuRoC MH01 [32], [31], [33], [34]. KITTI uses a visual ensemble and monocular ORB; other groups use inertial experts. Historical and current results remain separate.

B. Repeated evaluation of inertially guided SLAM

Fixed study. We evaluate all 11 EuRoC sequences and six TUM-VI room sequences [34], [33], with three fresh-process repetitions and both RGB-only and RGB–IMU expert modes: 102 executions in a fixed order. The visual checkpoint, camera input, loop-off configuration, admission thresholds and unit motion weight are fixed. Each execution retains first-publication camera poses across its complete input schedule. Scores use the original sealed outputs after uniform provenance verification; the evaluation does not repeat inference.

System benefit. The primary comparison pairs VIStream with the frozen visual trajectory from the same RGB–IMU execution, on identical GT-supported timestamps. A single whole-trajectory Sim(3) fit evaluates camera-center shape; it does not establish metric-scale accuracy. We first average the three repetitions within each sequence, then weight all 17 sequences equally. Across all 51 inertial executions, mean ATE decreases from 2.0764 to 0.7274 m, a 64.97% reduction. The paired hierarchical 95% interval for the mean difference

is $[-2.1605, -0.6574]$ m, using 10,000 resamples and the predeclared seed. Resampling uses sequences and paired repetitions, not individual frames.

The improvement appears in 46 of 51 executions and 15 of 17 sequence means. EuRoC mean ATE decreases from 2.6974 to 0.6322 m; TUM-VI room mean ATE decreases from 0.9379 to 0.9021 m. TUM-VI's interval crosses zero, indicating regime-dependent effects. The study extends the Oxford diagnostics across repeated indoor routes; historical exposure is retained, without a training-distribution separation claim.

Published metric motion. Post-hoc aggregation uses the original shared startup conversion g_0 on 117,874 common metric-published GT frames from all 51 runs. Original-adjacent translation/rotation RPE decreases from 0.0955 m/4.0124° to 0.0169 m/0.5868° (82.31%/85.38%), without further scale fitting. The 9,140 unavailable initialization frames remain in the 129,852-input coverage denominator. Shared estimated conversion makes this a conditional interface comparison, not isolated IMU attribution. SE(3) ATE decreases from 2.3709 to 1.0779 m, but its difference interval $[-2.4022, 0.0521]$ m crosses zero; TUM-VI reverses direction, 1.1078 to 2.3697 m. Full coverage, strata and residual-scale diagnostics appear in the supplement.

C. Sensor and scale-integration controls

Using a persistent inertial length unit. Once initialized, a calibrated visual–inertial estimate describes depths and translations in a common metric unit. We test how the interface uses this information by comparing the two recorded unit-weight branches from each of the same 51 inertial executions: preserving the first accepted scale, or accepting a new sparse-depth scale at each expert epoch. Visual outputs, expert returns, admission tests and four-way GT/visual/persistent/reset support are shared. This post-hoc comparison reuses the original trajectories and frozen evaluation; it does not run a new estimator.

Persistent scale reduces mean Sim(3) ATE from 1.2396 to 0.7274 m (41.32%), with a paired mean difference of -0.5121 m and hierarchical 95% interval $[-0.9932, -0.1406]$ m. It improves 43/51 runs and 16/17 sequence means. Rotation matrices are identical between these branches, localizing this effect to translation-scale integration. The analysis supports consistent use of initialized inertial geometry; it does not attribute the difference to new rotational estimation or certify absolute scale accuracy.

Separating sensor mode from interface policy. All 51 RGB/RGB–IMU pairs have identical visual poses and GT support. Under the same per-epoch interface, RGB-only/RGB–IMU ATE is 0.6111/1.2396 m; the paired difference is $+0.6284$ m, with 95% interval $[0.1914, 1.1265]$ m. Comparing persistent-scale inertial output to per-epoch RGB output changes both factors: its $+0.1163$ m difference, interval $[-0.1656, 0.3761]$ m, cannot isolate sensor benefit. Thus the system and scale gains above support distinct claims.

Gyro mechanism control. One fixed causal gyro integrator replaces rotation while retaining the original VI local translations, scale and admission rules; translations are reintegrated under the new orientation. All 51 recorded-input replays preserve the original full axis and GT-supported mask. Adjacent rotation RMSE falls from 0.7174 to 0.2962° (58.71%), whereas primary Sim(3) ATE rises from 0.7274 to 2.0329 m. This local/global distinction supports joint evaluation of orientation and trajectory, not a new primary model or an IMU-switch claim. The supplement reports the fixed protocol, all sequence results and nominal intervals; this full-native support differs from the metric-publication subset above.

D. Controlled treatment of expert revisions

Two controls are frozen across all 51 inertial runs before scoring. Returned increments replace available pre-guard expert deltas with successive-returned-pose differences and re-evaluate the original rate tests. Anchored components place returned poses through fixed anchors keyed conservatively by map/reset/creation counters, retaining original component masks and visual continuation. Scales, full schedules and warmup dependencies are shared; both original branches reproduce exactly. This is recorded-input CPU replay. The supplement specifies anchor initialization and recovery.

TABLE II

FIXED-INPUT REVISION CONTROLS: ALL 51 INERTIAL RUNS, 126,045 COMMON FRAMES. ONE GLOBAL SIM(3) FIT; RPE USES ORIGINAL ADJACENT PAIRS. EQUAL-SEQUENCE MEANS; BOLD: LOWEST UNROUNDED MEAN.

	ATE [m]	t-RPE [cm]	r-RPE [°]
Frozen visual	2.076	8.04	3.909
VIStream	**0.727**	**2.26**	**0.717**
Returned increments	0.735	2.37	0.737
Anchored components	1.210	5.71	1.256

Relative to returned-pose increments, VIStream's native increments reduce aggregate adjacent translation and rotation RPE by 4.86% and 2.66%, respectively. Returned and anchored ATE differences are $+0.0076$ and $+0.4827$ m; hierarchical intervals $[-0.0278, 0.0504]$ and $[-0.0476, 1.0924]$ m cross zero. Returned increments remove one rotation and 32 translation admissions; absolute corrections can exceed incremental rate bounds despite unchanged masks. This exploratory comparison supports retaining the current rule; it does not replace the stronger direct-expert comparison or isolate IMU benefit.

E. Oxford metric camera localization

ATE and relative-pose error assess global and local trajectories [35]. Native Sim(3) ATE [36] uses all GT-supported first outputs, including initialization and fallback. Metric SE(3) ATE fits no scale and requires a published metric gauge, excluding prefixes of 174, 231, and 102 frames. Expert comparisons share exactly these metric timestamps and GT support.

What the alignment measures. Physical camera calibration and evaluation alignment are distinct [37], [38]. Six-degree-of-freedom position alignment compares trajectories

TABLE III

KITTI ODOMETRY ATE RMSE (M). PUBLISHED HORIZONSTREAM COMPARISON ROWS ARE RETAINED; THE TWO LOCAL VISTREAM ROWS ARE HISTORICAL ORB LOOP-OFF AND AUXILIARY LOOP-ON ROUTES. BOLD MARKS PER-COLUMN MINIMA WITHIN THESE TWO LOCAL ROWS; TIES ARE BOTH BOLD.

Method	00	01	02	03	04	05	06	07	08	09	10	Avg.
Published HorizonStream comparison rows												
MASt3R-SLAM†	–	530.37	–	18.87	88.98	159.43	92.00	–	263.75	–	153.07	186.64
VGGT-SLAM†	–	607.16	–	169.83	13.12	–	–	–	–	–	211.82	250.48
COLMAP†	139.12	3.83	71.99	1.46	112.77	20.37	10.95	7.80	21.72	21.19	4.52	37.79
MASt3R-SfM†	–	463.52	–	15.80	41.44	150.39	136.14	71.69	–	176.36	69.50	140.60
DPVO†	113.11	16.60	113.01	2.46	0.98	59.34	55.91	19.30	110.63	74.55	13.71	52.69
VGGT-Long†	8.64	61.21	52.72	8.78	4.20	9.88	4.67	2.66	72.98	31.84	27.71	25.94
FastVGGT†	–	705.39	–	62.38	10.27	157.74	124.43	69.27	–	190.10	194.75	189.29
LoGeR†	54.98	36.57	36.20	4.27	1.62	33.41	11.78	13.33	22.92	17.89	8.06	21.91
LoGeR*†	26.19	41.26	32.21	5.02	1.62	22.65	5.49	5.04	21.96	9.03	9.44	16.35
CUT3R w/ refresh†	190.38	90.59	264.39	20.40	7.31	92.25	67.54	22.48	145.08	67.42	40.00	91.62
TTT3R w/ refresh†	119.94	99.59	238.07	16.83	3.98	36.38	47.20	11.62	107.33	86.96	33.58	72.86
LongStream†	92.55	46.01	134.70	3.81	1.95	84.69	23.12	14.93	62.07	85.61	21.48	51.90
LINGBOT-MAP†	30.80	64.74	82.29	2.49	0.85	16.55	6.27	8.92	39.32	17.99	7.96	25.29
HorizonStream (loop off)†	26.40	20.62	84.62	5.15	0.62	12.82	4.59	5.49	19.49	25.73	11.71	19.75
HorizonStream (loop on)†	13.91	20.62	69.43	5.15	0.62	6.86	6.50	2.67	19.49	23.86	11.71	16.44
VISTREAM (ORB loop off)	34.88	**16.10**	45.31	2.05	**0.27**	9.86	14.59	10.05	27.34	14.00	7.29	16.52
VISTREAM (ORB loop on)	**8.47**	**16.10**	**19.88**	**1.97**	0.62	**6.44**	**9.47**	**2.89**	**25.02**	**6.92**	**7.10**	**9.53**

† Official values are copied from HorizonStream. The two local rows are archived ORB routes: loop-off is 16.52 m and loop-on is 9.53 m. They are shown as system-level context, not as an isolated loop-switch ablation or a newly matched cross-paper leaderboard.

TABLE IV

CURRENT FIRST-OUTPUT ATE [M]. N: GT-SUPPORTED FRAMES. V: VISUAL; VI: VISTREAM. BOLD: MINIMUM ON IDENTICAL SUPPORT WITHIN EACH PROTOCOL. OBSERVATORY EXPERT SUPPORT DIFFERS AND IS NOT RANKED.

	Full native Sim(3)			Published metric SE(3)			
Sequence	N	V	VI	N	V	VI	Expert
Christ Church 02	3839	29.765	**2.591**	3666	30.722	**1.920**	2.045
Keble College 04	3838	31.268	**24.192**	3609	31.230	28.863	**1.341**
Observatory Quarter 01	3840	**28.020**	28.060	3738	39.218	**29.603**	–

without testing gravity-direction consistency. Local translation and rotation RPE use shared camera-relative endpoint transforms without interval fitting. Both global and local errors are needed to assess SLAM performance.

Calibrated motion reduces metric position ATE on all three matched supports (Table IV): 30.722 to 1.920 m on Christ Church, 31.230 to 28.863 m on Keble, and 39.218 to 29.603 m on Observatory. The largest gain occurs on Christ Church, which also improves in native shape. Observatory's native ATE slightly worsens, distinguishing metric position from scale-aligned shape. Against the stronger expert baseline, Keble remains worse (28.863 versus 1.341 m); Observatory's different-support expert cell is unavailable. The controlled same-expert comparison below isolates identical scheduled supports.

F. Local SLAM accuracy under camera motion

Shared intervals. We evaluate 553 approximately one-second intervals on 11,013 common metric frames, with contiguous input indices and disjoint interiors. This exploratory protocol was fixed after global results, before these diagnostics. GT rotation rate sums consecutive SO(3) angles; secant speed and time-normalized velocity differences describe motion. Each sequence's upper quartile is compared with its other intervals. Camera-relative RPE uses no window alignment or parameter fitting.

Motion benefit. VIStream lowers translation RPE in 96.0% of intervals relative to visual motion. Absolute RMSE reductions are larger in high-rotation strata (Table V); high-speed and high-acceleration strata also improve on all three sequences. Median translation RPE decreases from 2.140/1.361/3.795 to 0.052/0.051/0.116 m, showing gains beyond the largest-error intervals.

Same-expert events. Against the same expert plus com-

TABLE V

EXPLORATORY 1 S TRANSLATIONAL RPE RMSE [M]. HIGH: TOP QUARTILE OF GT CAMERA ROTATION RATE; LOWER: OTHER WINDOWS. V: FROZEN VISUAL; VI: VISTREAM; D: SAME EXPERT + SIMPLE COMPLETION. NO WINDOW FITTING. BOLD: LOWEST ORIGINAL VALUE PER ROW.

Motion stratum	N	V	VI	D
Christ Church 02				
High rotation	46	7.477	0.219	**0.074**
Lower rotation	138	2.840	0.255	**0.178**
Keble College 04				
High rotation	46	2.574	**0.471**	1.357
Lower rotation	135	1.494	0.586	**0.287**
Observatory Quarter 01				
High rotation	47	18.691	8.535	**0.107**
Lower rotation	141	5.133	2.229	**0.115**

pletion, Christ Church's high-acceleration translation RMSE decreases by 3.08% (0.08243 to 0.07989 m; 33/46 wins); mean and median also improve. Keble high-rotation translation RMSE decreases from 1.357 to 0.471 m, with 13/46 wins and 88.4% of the direct baseline's squared error concentrated in one interval. These are event-dependent translation gains; the control retains lower rotational RMSE. Supplementary Fig. S2 and its accompanying diagnostics retain all 553 intervals, both pose-error components, quantiles and the matched-event sensitivity checks.

G. Same-expert trajectory comparison

Direct uses the same returned expert camera poses, calibration and fixed metric scale, with visual-increment gap completion and the original recovery anchor.

Repeated-cohort benefit. Across all 51 inertial executions (Sec. IV-B), sharing expert returns, frozen geometry, calibration and metric-published GT support, equal-sequence mean SE(3) ATE decreases from Direct's 3.5149 to 1.0779 m (69.33%); original-adjacent translation RPE decreases from 0.187749 to 0.016901 m (91.00%). Matching caller admission retains a 69.38% ATE reduction. These post-hoc gains describe the fixed 17-sequence cohort; the supplement retains every sequence mean and metric stratum.

Oxford scheduled-support readout. Table VI retains the separate 306/301/312-frame schedules. Independent rigid camera-center fits share these timestamps; RPE uses consecutive scheduled pairs, including gaps. These fits never affect auxiliary maps. Corrected pose admission restores 115/445/618 safe poses while retaining tracking, initialization, inlier, pose and provenance checks.

Christ Church ATE improves 5.12%; Keble's 26.13% translation-RPE RMSE reduction is event-concentrated: VIStream wins 123/300 approximately 0.6 s pairs, and one Direct pair contributes 86.4% of Direct's squared error. Removing it post hoc gives 0.198/0.396 m (Direct/VIStream). Table VI retains all positional and rotational endpoints; this RMSE reduction is interpreted at the event level.

Direct covers 3,666/3,609 Christ Church/Keble metric frames. Observatory misses 20/3,738, including 1/312 scheduled frames; both methods complete them, giving equal coverage. No motion gates trigger across 9,486 admitted increments; their removal changes no trajectory, and no initialized map transition occurs. These runs therefore do not establish benefits from either mechanism. Readouts retain their original admission, absolute-pose and fallback histories, including on the direct-valid intersection. Equation 4 characterizes how offsets can persist into a later coherent interval; it does not identify the onset or events responsible for these Oxford errors. Supplementary diagnostics retain support and endpoint errors.

H. Reliability and failure controls

Current validity checks. Controlled replay perturbs recorded returns on unchanged Christ Church metric support. Table VII shares one clean-trajectory rigid transform across perturbations. Rate checks limit a large rotational jump and a gross translation spike. An admitted 2° bias remains damaging. These test the adapter on recorded outputs, not estimator responses to modified sensors.

The separate bounded absolute-correction prototype retains its six-cap saturation tests. Full $SE(3)$ boundary and missing-interval controls can improve missing-frame rotation while worsening endpoint position; they establish no universal drift bound.

I. Cross-dataset SLAM trajectory context

Table VIII preserves historical batch measurements with paired frame support and independent Sim(3) alignment; these do not evaluate Eq. 3.

Table III retains 15 published HorizonStream comparison rows and our archived ORB loop-off/on routes (16.52/9.53 m). Expert availability and fallback can differ, precluding an isolated switch ablation. Published and local rows retain their respective protocols.

Fig. 3 shows three trajectories. SE(3) and relative-motion checks on 14 historical inertial candidates retain position gains, but rotation RPE worsens on Oxford and MH01. A separate Oxford implementation achieves 1.7527 m macro ATE on ten sequences. Four original global-cloud exports have differing sampling and no paired visual controls; they remain downstream diagnostics.

Historical completion controls. With fixed expert and overlapping outputs, boundary completion lowers ATE from 7.3503 to 3.2687 m on Oxford and from 0.0806 to 0.0162 m on TUM-VI. Oxford endpoint/adaptive completion gives independent Sim(3) ATE 2.5086/3.2687 m and shared-overlap-transform prefix RMSE 4.9065/9.1781 m.

Raw-IMU-only KITTI gives 58.5469 versus 58.5485 m for its full-frame baseline; it does not isolate the differently sampled ensemble's IMU benefit.

Holding positions and observed rotations fixed, overlap-anchored missing rotations give RPE 2.3828/0.4878/0.1902 degrees on Oxford/TUM-VI/MH01. These controls test historical completion.

J. Execution cost and scope of transfer

On a shared H20, the instrumented implementation produces 1.527/1.531/1.557 first outputs/s. The approximately

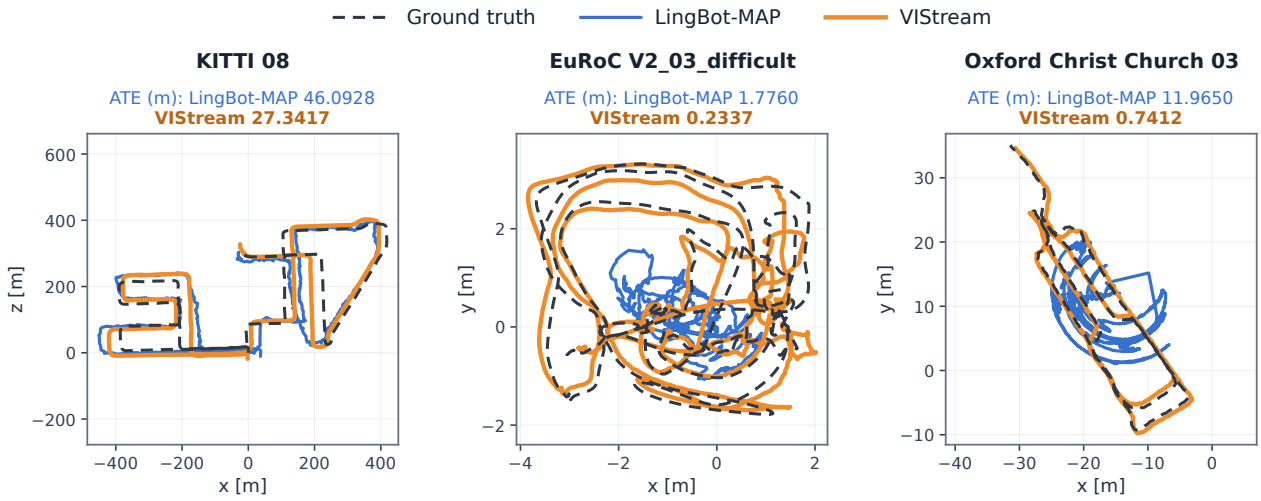


Fig. 3. Batch trajectories, independently Sim(3)-aligned on matched camera-center frames. Christ Church 03 has the lowest LingBot-Map ATE among ten retained Oxford sequences. V2_03 is outside the MH01 aggregate; KITTI 08 uses archived ORB-off fallback.

TABLE VI

FIRST-OUTPUT TRAJECTORY COMPARISON ON THE ORIGINAL FIXED COMMON-FRAME SCHEDULE. D: THE SAME EXPERT'S DIRECT ABSOLUTE CAMERA POSES PLUS SIMPLE VISUAL COMPLETION; VI: VISTREAM. CALIBRATION, METRIC GAUGE, AND SELECTED TIMESTAMPS ARE SHARED. ATE USES INDEPENDENT SE(3) ALIGNMENT. RPE IS MEASURED BETWEEN ADJACENT ENTRIES OF THE FIXED SCHEDULE, INCLUDING GAPS; IT IS NOT A FIXED 1 S OR RAW-FRAME INTERVAL. LOWER IS BETTER. BOLD MARKS THE SMALLER ORIGINAL VALUE IN EACH D/VI PAIR.

		ATE RMSE [m]		Scheduled T-RPE RMSE [m]		Scheduled R-RPE RMSE [°]	
Sequence	N	D	VI	D	VI	D	VI
Christ Church 02	306	2.046	**1.942**	**0.124**	0.200	**0.156**	2.109
Keble College 04	301	**1.349**	28.998	0.537	**0.397**	**0.705**	4.227
Observatory Quarter 01	312	**1.334**	29.743	**0.290**	3.148	**0.478**	7.101

TABLE VII

CURRENT-METHOD RECORDED-OUTPUT FAULT CONTROLS. ATE [M] USES ONE FIXED CLEAN SE(3) TRANSFORM ON ALL 3,666 METRIC FRAMES. RATE-CHECK REMOVAL CHANGES NO OTHER ADMISSION RULES. LOWER IS BETTER; TIES ARE BOTH BOLD.

Condition	Normal checks	Rate checks off
Clean	**1.920**	**1.920**
Dropout / reset	**6.985**	**6.985**
90° perturbation	**5.323**	76.989
Translation spike ×1000	**1.921**	26.634
Admitted 2° bias	**28.480**	**28.480**
Nonrigid perturbation	**2.542**	**2.542**

TABLE VIII

HISTORICAL BATCH MACRO SIM(3) ATE [M]. REDUCTION USES THE LOCAL FROZEN VISUAL SOURCE; KITTI INCLUDES AN UPSTREAM ENSEMBLE.

Dataset group	Frozen visual	Adapted	Reduction
KITTI 00–10 (ORB off)	32.4867	**16.5203**	49.1%
Oxford, 10 sequences	32.4762	**3.2687**	89.9%
TUM-VI rooms 2/4/6	0.9954	**0.0162**	98.4%
EuRoC MH01	2.5425	**0.0249**	99.0%

193 s Oxford sequences take 2,515/2,508/2,467 s from first dispatch to last output. This ten-branch instrument supports sequential processing but does not match the approximately 20 Hz input.

Transfer and motion coverage. Frozen reuse needs no fine-tuning; held-out transfer remains unestablished. LingBot-Map training includes road/flight data, variable strides and synthesized turns [4]. Our GT-derived speeds, 0.660–1.567 m/s, overlap its 0.3–1.8 m/s synthetic-subset range, which does not characterize all training motion.

Oxford raw gyroscope-magnitude 95th percentiles are 36.23/41.93/37.38 degrees/s across the RGB spans. Accelerometers measure specific force, not gravity-free acceleration. Establishing motion-distribution shift requires actual training samples and timestamps with matched units and differentiation.

K. *Auxiliary fixed-geometry readout*

The 306/301/312-frame Oxford schedules share frozen depths, scale, masks and a GT-camera anchor, without trajectory fitting or ICP. Completeness improves over frozen visual motion on all three; the direct expert has higher F10 throughout. Supplementary Tables S1–S2 and Fig. S1 retain all metrics and maps as auxiliary consequences of SLAM poses.

V. Discussion and Conclusion

VIStream provides calibrated SLAM motion integration with frozen upstream models. On the fixed 17-sequence cohort, ATE decreases by 64.97% versus frozen visual motion (Sim(3)) and 69.33% versus same-expert direct poses with completion (SE(3)). Within the same inertial executions, persistent scale lowers Sim(3) ATE by 41.32% versus epoch-local recalibration.

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